

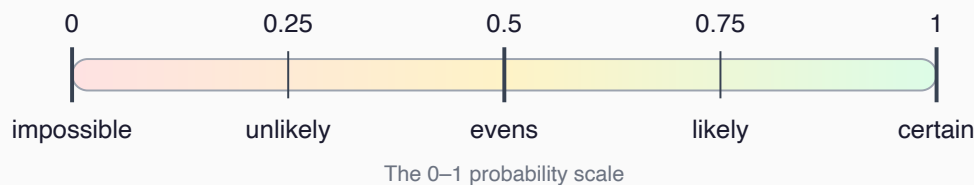
- RAG Learning

5.1 Probability

Pearson Edexcel GCSE Mathematics 1MA1 — Probability — spec refs P1–P9 — Papers 1 (non-calculator), 2 and 3 (calculator)

5.1.1 The 0–1 Scale and Theoretical Probability (P3, P4, P7)

A **probability** is a number from 0 (**impossible**) to 1 (**certain**) that measures how likely an event is. Give probabilities as **fractions, decimals or percentages**. Answers written as a ratio ("3 : 10") or in words ("3 out of 10") are not accepted by Edexcel mark schemes.



Equally Likely Outcomes

If every outcome is **equally likely** (a fair dice, a random choice):

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{total number of possible outcomes}}$$

Worked Example: A bag contains 4 red, 7 blue and 9 green counters. One counter is taken at random. Find $P(\text{blue})$ and $P(\text{not red})$.

$$\text{Total} = 20. \quad P(\text{blue}) = \frac{7}{20}. \quad P(\text{not red}) = \frac{16}{20} = \frac{4}{5}$$

Probabilities That Sum to 1 (P4)

The probabilities of an **exhaustive** set of **mutually exclusive** outcomes add up to 1, so

$$P(\text{not } A) = 1 - P(A)$$

Worked Example: A biased spinner lands on red, blue, green or yellow. $P(\text{red}) = 0.25$, $P(\text{blue}) = 0.4$ and $P(\text{green}) = 0.15$. Find $P(\text{yellow})$.

$$P(\text{yellow}) = 1 - (0.25 + 0.4 + 0.15) = 1 - 0.8 = \mathbf{0.2}$$

Worked Example: A four-sided spinner has $P(A) = 2x$, $P(B) = x$, $P(C) = 0.35$ and $P(D) = x + 0.05$. Find x .

$$2x + x + 0.35 + x + 0.05 = 1 \Rightarrow 4x + 0.4 = 1 \Rightarrow 4x = 0.6 \Rightarrow x = \mathbf{0.15}$$

Exam Tip: Sense-check every answer: a probability must lie between 0 and 1. If you get 1.3 or a negative value, look again. You do not need to simplify fractions unless asked, but an incorrect "simplification" can lose the accuracy mark.

5.1.2 Experiments, Relative Frequency and Expected Results (P1, P2, P3, P5)

Relative frequency = $\frac{\text{number of times the outcome happened}}{\text{total number of trials}}$. It is an **estimate** of the probability, found by experiment rather than theory.

By P5, as the number of trials grows, the relative frequency of an unbiased experiment settles towards the **theoretical probability**. That is why results from more trials give a **more reliable** estimate — and why a few odd results do not prove something is biased.

Worked Example: A coin is thrown 250 times and lands on heads 140 times. Estimate $P(\text{heads})$ and comment on whether the coin is fair.

Relative frequency $= \frac{140}{250} = 0.56$. This is higher than 0.5, so the coin *might* be biased towards heads, but 250 throws cannot prove it. Throwing it many more times would give a more reliable estimate.

Fairness and Randomness (P2)

A **fair** (unbiased) dice, coin or spinner has all outcomes equally likely. Selecting “at random” means every item has the same chance. Even a fair dice rolled 30 times will rarely give exactly 5 of each score — randomness produces variation.

Expected Outcomes (P2, P3)

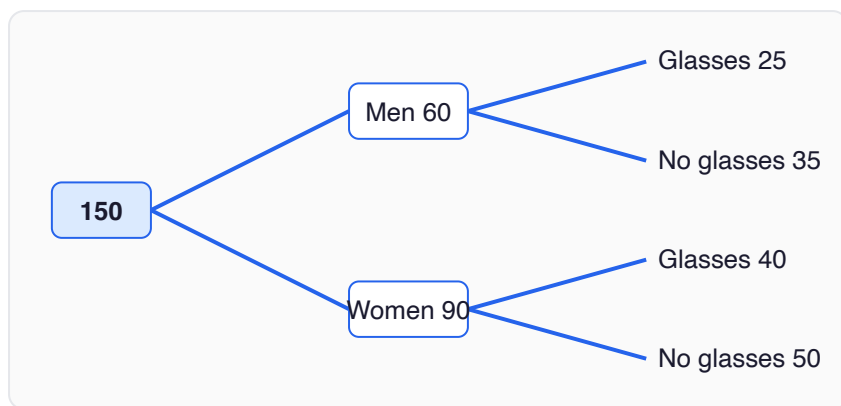
$$\text{Expected number of times} = P(\text{outcome}) \times \text{number of trials}$$

Worked Example: A game costs £1.50 to play. The probability of winning the £8 prize is 0.1. The game is played 300 times. Work out the expected profit for the organiser.

Expected wins $= 0.1 \times 300 = 30$. Takings $= 300 \times £1.50 = £450$. Prizes $= 30 \times £8 = £240$.
Expected profit $= £450 - £240 = \textbf{£210}$

Frequency Trees (P1)

A frequency tree splits a group in stages and records **frequencies** (counts) on the branches, not probabilities. The numbers at each split must add up to the number before it.



From the tree: $P(\text{a woman who wears glasses}) = \frac{40}{150} = \frac{4}{15}$, and $35 + 50 = 85$ people do not wear glasses.

5.1.3 Systematic Listing and Possibility Spaces (P6, P7)

A **possibility space** (sample space) shows every possible outcome. Work **systematically**: keep the first choice fixed while you run through every option for the second, then move on.

Worked Example: Ella chooses one drink (tea or juice) and one snack (apple, crisps or cake). List all the possible combinations.

tea–apple, tea–crisps, tea–cake, juice–apple, juice–crisps, juice–cake (6 combinations)

Two-Dice Possibility Space

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10

5	6	7	8	9	10	11
6	7	8	9	10	11	12

All 36 cells are equally likely. $P(\text{total} = 8) = \frac{5}{36}$ and $P(\text{total} \geq 10) = \frac{6}{36} = \frac{1}{6}$.

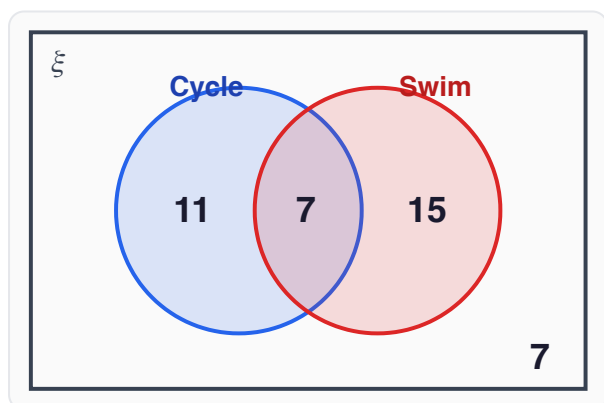
Worked Example: Two fair dice are rolled and the scores are multiplied. Find the probability that the product is odd.

The product is odd only when both scores are odd: $3 \times 3 = 9$ of the 36 cells.

$$P(\text{odd product}) = \frac{9}{36} = \frac{1}{4}$$

5.1.4 Venn Diagrams and Sets (P6)

A Venn diagram sorts items into overlapping sets inside a rectangle that holds everything (the **universal set** ξ). **Start with the overlap**, then subtract to fill each “only” region, and finally the outside.



Worked Example: 40 students: 18 cycle, 22 swim and 7 do both.

Overlap = 7. Cycle only = $18 - 7 = 11$.

Swim only = $22 - 7 = 15$.

Neither = $40 - (11 + 7 + 15) = 7$.

$$P(\text{swims but does not cycle}) = \frac{15}{40} = \frac{3}{8}$$

Set Notation

Edexcel questions on Venn diagrams often use set notation, so learn these symbols:

Notation	Meaning	In the diagram above
ξ	Universal set — everything being considered	all 40 students
$A \cap B$	Intersection: in A and in B	7
$A \cup B$	Union: in A or B or both	$11 + 7 + 15 = 33$
A'	Complement: not in A	do not cycle: $15 + 7 = 22$
$A \cap B'$	In A but not in B	cycle only: 11
$(A \cup B)'$	In neither set	7

Worked Example: $\xi = \{1, 2, 3, \dots, 10\}$, $A = \{\text{even numbers}\}$ and $B = \{\text{factors of } 10\}$. List $A \cap B$ and $A \cup B$.

$$A = \{2, 4, 6, 8, 10\}, \quad B = \{1, 2, 5, 10\}$$

$$A \cap B = \{2, 10\}, \quad A \cup B = \{1, 2, 4, 5, 6, 8, 10\}$$

5.1.5 Combined Events: Adding and Multiplying (P4, P8)

Mutually exclusive events cannot happen together. **Independent** events do not influence each other: knowing one happened does not change the probability of the other.

OR (mutually exclusive events): $P(A \text{ or } B) = P(A) + P(B)$

AND (independent events): $P(A \text{ and } B) = P(A) \times P(B)$

At least one: $P(\text{at least one}) = 1 - P(\text{none})$

Worked Example: The probability of rain on Saturday is 0.4 and on Sunday is 0.3. Assume the days are independent. Find the probability that (a) it rains on both days (b) it rains on neither day.

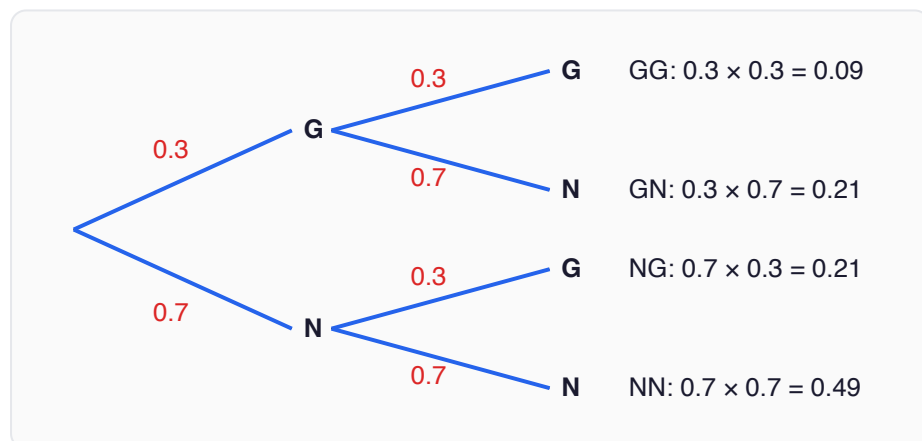
(a) $0.4 \times 0.3 = \mathbf{0.12}$ (b) $0.6 \times 0.7 = \mathbf{0.42}$

Worked Example: Three fair coins are thrown. Find $P(\text{at least one head})$.

$P(\text{no heads}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$, so $P(\text{at least one head}) = 1 - \frac{1}{8} = \frac{7}{8}$

5.1.6 Tree Diagrams, Independent and Dependent Events (P6, P8)

A tree diagram shows events in order. **Multiply along** a path to get the probability of that combined outcome; **add** the results of different paths for "or". The branches leaving any point must add to 1, and all end results together add to 1.



A spinner lands on green (G) with probability 0.3 and is spun twice. $P(\text{exactly one green}) = 0.21 + 0.21 = 0.42$. Check: $0.09 + 0.21 + 0.21 + 0.49 = 1 \checkmark$

Without Replacement (Dependent Events)

When an item is **replaced**, the second-stage probabilities are the same as the first (independent events). When it is **not replaced**, the events are **dependent**: the total drops by 1, and so does the number of the type already taken.

Worked Example: A bag holds 4 red and 6 blue counters. Two are taken at random without replacement. Find $P(\text{one of each colour})$.

$$P(R, B) = \frac{4}{10} \times \frac{6}{9} = \frac{24}{90} \quad P(B, R) = \frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$$

$$P(\text{one of each}) = \frac{48}{90} = \frac{8}{15}$$

Worked Example: Higher A bag contains n sweets, 3 of which are orange. Two are taken without replacement and $P(\text{both orange}) = \frac{1}{12}$. Find n .

$$\frac{3}{n} \times \frac{2}{n-1} = \frac{1}{12} \Rightarrow n(n-1) = 72 \Rightarrow n^2 - n - 72 = 0$$

$$(n-9)(n+8) = 0, \text{ so } n = 9 \text{ (reject } -8\text{)}. \text{ This needs Higher-tier quadratic solving (A18).}$$

Exam Tip: P8 asks you to “know the underlying assumptions”: multiplying probabilities assumes **independence**, and using a relative frequency assumes the experiment is a fair estimate. Look for “without replacement”, “takes two” or “eats” — then the second denominator must go down by 1.

5.1.7 Conditional Probability (P9) Higher

A **conditional probability** is the probability of A *given that* B has happened, written $P(A | B)$. Knowing that B happened **shrinks** the set of possible outcomes to those in B .

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

The 1MA1 spec asks you to work with conditional probability through **expected frequencies** in two-way tables, tree diagrams and Venn diagrams.

From a Two-Way Table

	Walk	Bus	Total
Year 10	27	18	45
Year 11	22	33	55
Total	49	51	100

Worked Example: A student who travels by bus is chosen at random. Find the probability that they are in Year 11.

Use only the "Bus" column: 51 students, 33 of them in Year 11.

$$P(\text{Year 11} \mid \text{bus}) = \frac{33}{51} = \frac{11}{17} \quad (\text{compare } P(\text{Year 11}) = \frac{55}{100})$$

From a Tree Diagram

Worked Example: $P(\text{snow}) = 0.2$. If it snows, $P(\text{school closes}) = 0.6$; if it does not snow, $P(\text{school closes}) = 0.05$. Given that the school is closed, find the probability that it snowed.

$$P(\text{closed}) = 0.2 \times 0.6 + 0.8 \times 0.05 = 0.12 + 0.04 = 0.16$$

$$P(\text{snow} \mid \text{closed}) = \frac{0.12}{0.16} = \frac{3}{4}$$

From a Venn Diagram

Worked Example: $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$. Find $P(A \cap B)$, $P((A \cup B)')$ and $P(A \mid B)$.

$$P(A \cap B) = 0.4 + 0.5 - 0.7 = \mathbf{0.2}, \quad P((A \cup B)') = 1 - 0.7 = \mathbf{0.3}$$

$$P(A \mid B) = \frac{0.2}{0.5} = \mathbf{0.4} \quad (\text{equal to } P(A), \text{ so here } A \text{ and } B \text{ are independent})$$

Expected Frequencies

Turning probabilities into counts out of a convenient total makes conditional questions much easier. In the snow example, out of 100 days: 20 snowy days give 12 closures and 80 other days give 4 closures, so 12 of the 16 closures were on snowy days — $\frac{12}{16} = \frac{3}{4}$ again.

Exam Tip: Spot the trigger words: "given that", "of those who", "a student who ... is chosen". The denominator becomes the size of that **given group**, not the grand total. If $P(A | B) = P(A)$, the events are independent.

Summary

Idea	Rule
Complement	$P(\text{not } A) = 1 - P(A)$
Relative frequency	frequency $\div$ number of trials
Expected frequency	probability $\times$ number of trials
Mutually exclusive (or)	$P(A) + P(B)$
Independent (and)	$P(A) \times P(B)$
At least one	$1 - P(\text{none})$
Conditional Higher	$P(A B) = \frac{P(A \cap B)}{P(B)}$